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Supporting Information

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Theoretical Considerations on the Electroreduction of CO to C_2 Species on Cu(100) Electrodes**

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Herein we provide the details of the calculations and describe the corrections added to the DFT-calculated reaction energies in order to transform them into Gibbs energies, and the details of the transition state calculations. All energies are provided in eV.

1. Computational Details

All adsorption energies were calculated by means of DFT simulations using the Vienna Ab-initio Simulation Package (VASP)[1] with the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional.[2] Limitations in the energy of formation of CO (g) have been found for PBE and PW91[3]; besides, we found non-negligible discrepancies in the energies of formation of several C_2 species compared to experiments. Therefore, we systematically correct the energies of these molecules in the way shown in the Supplementary Information. The Cu(100) surfaces were modeled with 3×2 supercells with a lattice constant of 3.64 Å, and four metal layers were used. During structure optimization, the adsorbates and the two topmost layers were allowed to relax in all directions while the 2 bottom layers were fixed at the optimized bulk positions. The relaxations of the atoms were carried out with the quasi-Newton minimization scheme, until the maximum force on any atom was below 0.05 eV $Å^{-1}$. Ionic cores were described by Projector Augmented Wave (PAW) potentials.[4] The vertical separation between successive slabs was in all cases more than 16 Å and dipole corrections were applied. The search for optimal geometries was performed using a cut-off for the plane-wave basis set of 400 eV. The Brillouin zones of all surfaces were sampled with 6×8×1 Monkhorst-Pack grids.[5] The Fermi level of surfaces with and without adsorbed species was smeared by the Methfessel-Paxton approach[6] with a Gaussian width of 0.2 eV, and all energies were extrapolated to T = 0 K. Molecules were calculated with a width of 0.001 eV and Gaussian smearing was used instead. The DFT-calculated energies are converted into free energies in the following way:

$$\Delta G = \Delta E_{DFT} + \Delta ZPE - T\Delta S \quad \text{(S.1)}$$

The zero-point energies (ZPE) of adsorbates and molecules were calculated from the vibrational frequencies obtained within the harmonic oscillator approximation. Vibrational contributions to the entropy were considered for adsorbed species and total entropies for molecules were taken from standard thermodynamic tables at 298.15K and 1 atm.[7]

2. Computational Hydrogen Electrode

This approach capitalizes on the fact that protons and electrons are in equilibrium with hydrogen at 0 V, all pH values and 1 atm of pressure, if the reaction (S.2) is used as the reference potential at any pH:

$$\tfrac{1}{2} H_{2(g)} \leftrightarrow H^{+}_{(aq)} + e^{-} \quad \text{(S.2)}$$

At these conditions, the chemical potential (μ) of ½H_2 (g) is equivalent to that of a proton-electron pair in aqueous solution. Thus, the energetics of H_2 (g) rather than that of *(H^++e^-)* is used as electrochemical reference. The effect of an external potential over this equilibrium can also be included as a term *eU* in the following way, where *U* is the electrode potential vs. the RHE:

$$\tfrac{1}{2}\mu\left(H_2\right) - eU = \mu\left(H^+ + e^-\right) \tag{S.3}$$

3. Gas-phase Corrections

We estimate the free energies of gas-phase molecules with a formula analogous to Eq (S.1):

$$G = E_{DFT} + ZPE - TS \tag{S.4}$$

The *ZPE* and *TS* at 298.15 K used in this study for various gas-phase species are provided in Table S.1.

Table S.1. Zero-point energies and entropic corrections at 298.15 K for the gas-phase molecules included in this study. Data for graphene (C (s)) are also provided. The *TS* terms are taken from Ref.[7].

species	*ZPE*	*TS*
C (s)	0.13	0.02
CO	0.14	0.61
C_2H_2	0.72	0.62
C_2H_4	1.36	0.68
C_2H_6	1.98	0.71
H_2	0.27	0.40
H_2O	0.57	0.58
O_2	0.10	0.63
C_2H_5OH	2.11	0.87
CH_3CHO	1.47	0.82
C_2H_4O	1.52	0.75

As noted earlier by Peterson et al,[3b] the exchange-correlation functional PBE[2] has a limitation in its description of the standard formation energy (ΔG^0, given by the difference between the Gibbs energies of products and reactants) of CO (g). On the other hand, we found that this xc-functional also accurately describes the formation energies of gas-phase molecules with C-C bonds. Those formation energies are provided in Table S.2. The reference state for C is graphene, and H_2 and O_2 are taken in gas phase at 1 atm. We made sure that the formation energy of O_2 is correctly described by using the methodology described in Ref.[8].

Table S.2. Formation energies of the gas phase of various substances included in this study at 298.15 K and 1 atm. The experimental data are taken from Ref.[7].

GAS-PHASE REACTIONS	ΔG^0_{DFT}	ΔG^0_{EXP}	$\left\|\Delta G^0_{EXP} - \Delta G^0_{DFT}\right\|$
$C + \frac{1}{2}O_2 \rightarrow CO$	-1.18	-1.42	0.24
$2C + 3H_2 \rightarrow C_2H_6$	-0.30	-0.33	0.03
$2C + 2H_2 \rightarrow C_2H_4$	0.74	0.71	0.03
$2C + H_2 \rightarrow C_2H_2$	2.29	2.18	0.12
$2C + 3H_2 + \frac{1}{2}O_2 \rightarrow C_2H_5OH$	-1.74	-1.74	0.00
$2C + 2H_2 + \frac{1}{2}O_2 \rightarrow CH_3CHO$	-1.47	-1.38	0.09
$2C + 2H_2 + \frac{1}{2}O_2 \rightarrow C_2H_4O$	-0.06	-0.13	0.08

The mean absolute error (MAE) of the values in the fourth column of Table S.2 is 0.09 eV before correcting the energy of CO by 0.24 eV and 0.05 eV after the correction. Therefore, that number was used to correct for the inaccuracies in the description of CO and no additional corrections were needed for the other molecules in this study.

4. Liquid-phase Corrections

Water, ethanol and acetaldehyde are main products of the CO reduction on Cu(100) and they are formed in liquid phase. Note that it is normally difficult to estimate the liquid-phase free energies of substances within standard DFT. However, there is a workaround based on the free energies of formation of the liquid and gas phases of a given substance. The method is explained taking water as an example: the free energies of formation at 298.15 K of the liquid and gas phases are -2.4578 and -2.3691 eV, respectively, and the reactions are provided below:

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \tag{S.5}$$

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(g) \tag{S.6}$$

This means that the reactants are in both cases hydrogen and oxygen in gas phase. Then the following equality applies:

$$\Delta G^l_{H_2O} - \Delta G^g_{H_2O} = -0.0887 \tag{S.7}$$

Given that the free energies of Eqns (S.5) and (S.6) are $\Delta G^l_{H_2O} = G^l_{H_2O} - G_{H_2} - \frac{1}{2}G_{O_2}$ and $\Delta G^g_{H_2O} = G^g_{H_2O} - G_{H_2} - \frac{1}{2}G_{O_2}$, Eqn (S.7) can be simplified to:

$$G^l_{H_2O} = G^g_{H_2O} - 0.0887 \tag{S.8}$$

Norskov and coworkers[9] normally incorporate the difference between free energies into the entropy of gas-phase water, so that we have the following expression:

$$G^l_{H_2O} = E^{DFT}_{H_2O} + ZPE - 0.6722 \tag{S.9}$$

The same method was applied to describe liquid ethanol and acetaldehyde, for which we used TS corrections of 0.94 and 0.87 eV, i.e. 0.07 and 0.05 eV extra with respect to the values in Table 1, respectively.

5. Solvation Corrections for Adsorbates

In order to determine the contribution of solvation to the free energies, we used √3×√3 Cu(111) cells covered with and without water and *OH, *COH and *CO. The difference between the adsorption energies with and without water gives a good estimation of the magnitude of solvation, which is inevitably present in aqueous environments and should, therefore, be included in the free energies.[3b, 9-10] We find that *OH in a half-dissociated water layer is stabilized by 0.58 eV, in line with previous estimations of 0.50 eV.[3b, 10] Moreover, *COH was stabilized by 0.38 eV, while *CO was stabilized by 0.07 eV, consistent with the value of 0.1 eV.[3b] For consistency and to facilitate the direct comparison with similar works, we corrected the adsorption energies of *OH by -0.50 eV and those of *CO and CO-, CHO-containing adsorbates were corrected by -0.1 eV. However, we used our calculated correction of -0.38 eV for *ROH where R is a hydrocarbon chain, as the recommended in Ref[3b] originates from the assumption that the solvation of *OOH on Pt(111) is similar to that of *ROH on Cu. The last column of Table S.3 contains the kinds of solvation corrections added to each adsorbed species, where I stands for CO-, CHO-containing adsorbates, II for *ROH, and III for *OH.

6. Formation Energies of Reaction Intermediates

The adsorption energies are calculated with respect to CO and protons and electrons:

$$* + 2CO_{(g)} + n\left(H^+ + e^-\right) \rightarrow *C_2O_{2-x}H_{n-2x} + xH_2O_{(l)} \tag{S.10}$$

$$\Delta G_{*C_2O_{2-x}H_{n-2x}} = G_{*C_2O_{2-x}H_{n-2x}} + xG_{H_2O} - G_* - 2G_{CO} - \frac{n}{2}G_{H_2} + neU \tag{S.11}$$

On the other hand, Table S.3 contains the formation energies of the adsorbed states with respect to the clean 3×2 Cu(100) surface, graphene, H_2 and O_2. The preferred adsorption sites of C_2 species on the (100) surface were found by considering several configurations on the surface and the most stable one was chosen in each particular case. The Table also contains the ZPEs and the vibrational entropies of the adsorbed species, shown by Peterson et al[3b] to be important for an accurate determination of the potential-limiting step. As an example, we show the way of using the numbers in Tables S.2 and S.3 to compute the free energy of the first step in the pathways shown in Figure 1 in the main text. That step is the following:

$$2CO + \left(H^+ + e^-\right) + * \rightarrow *CO - COH \tag{S.12}$$

On the other hand, the formation energy of *CO-COH is calculated as:

$$2C + O_2 + \tfrac{1}{2}H_2 + * \rightarrow *CO - COH \tag{S.13}$$

Table S.3. Zero-point energies, vibrational entropy corrections and formation energies for all adsorbates considered in this study. The energies contain all the corrections mentioned in the text related to gas and liquid phases and solvation.

Description	Adsorbate	ZPE	TS^{vib}	ΔG^{f}	Solvation
Species with 2C and 2O atoms	2*CO	0.36	0.32	-3.24	I×2
	*C_2O_2	0.41	0.17	-2.28	I×3
	*C_2O_2 (TS)	0.37	0.22	-2.04	I×3
	*CO-COH	0.70	0.24	-2.43	I, II
	*COCHO	0.68	0.25	-2.28	I
	*OCHCHO	1.03	0.22	-2.97	I
	*OHCCOH	1.04	0.25	-2.89	II×2
	*OHCHCO	1.00	0.24	-2.21	I, II
	*OHCCHO	1.03	0.24	-2.78	I, II
	*OCH_2CH_2O	1.68	0.23	-2.86	-
	*OHCHCHO	1.31	0.29	-3.04	I, II
	*OCH_2CHO	1.30	0.32	-2.78	I
	*OCH_2CH_2OH	1.93	0.35	-3.15	II
Species with 2C and 1 O atoms	*CCO	0.34	0.18	-1.18	I
	*CHCO	0.61	0.21	-0.86	I
	*CCOH	0.59	0.19	-0.29	II
	*CCHO	0.60	0.17	-0.20	I
	*CHCOH	0.90	0.18	-0.80	II
	*CHCHO	0.91	0.18	-1.02	II
	*CCH_2O	0.89	0.17	0.63	-
	*CCHOH	0.92	0.22	-0.90	II
	*CH_2CHO	1.19	0.26	-1.26	II
	*$CHCH_2O$	1.20	0.15	-0.18	-
	*CHCHOH	1.19	0.26	-0.77	II
	*CH_2CH_2O	1.49	0.21	-0.84	-
	*CH_2CHOH	1.50	0.35	-1.12	II
	*$CHCH_2OH$	1.20	0.28	-0.68	II
	*CH_2CH_2OH	1.79	0.28	-1.07	II
	*CH_3CH_2O	1.82	0.31	-1.57	-
Species with only 2C and H atoms	*CC	0.21	0.08	2.29	-
	*CCH	0.46	0.12	1.71	-
	*CHCH	0.77	0.11	1.43	-
	*CCH_2	0.77	0.13	1.58	-
	*CH_2CH	1.07	0.21	1.44	-
	*$CHCH_3$	1.33	0.21	1.54	-
	*CH_2CH_3	1.64	0.25	0.84	-
O-related species	*O	0.06	0.05	-2.00	-
	*OH	0.33	0.11	-2.80	III

According to Table S.3, this formation energy (denoted as $\Delta G^{f}_{*CO-COH}$) is -2.43 eV. Table S.2 shows that the calculated and experimental formation energy of CO, ΔG^{f}_{CO}, is -1.42 eV. Therefore, it can be shown that the free energy of the first step of the CO reduction, ΔG_1, is:

$$\Delta G_1 = \Delta G^{f}_{*CO-COH} - 2\Delta G^{f}_{CO} = -2.43 + 2\times 1.42 = 0.40\ eV \tag{S.14}$$

7. Neutral and Charged CO Dimers in Gas Phase

Figure S.1 shows that both the calculated association enthalpy (ΔH) and its corresponding activation energy (E_a) for CO in gas phase to form C_2O_2 are high, 1.42 and 4.48 eV, which makes CO dimerization a rather unlikely process. Note that the instability of C_2O_2 has long been known.[11] Conversely, the charged species $C_2O_2^+$, $C_2O_2^-$ and $C_2O_2^{2-}$ are well known to be stable.[12] To the best of our knowledge, there are no reported thermodynamic for C_2O_2 (such as gas-phase entropy and formation energy), this being the reason why we have had to assume that its description is approximately as inaccurate as that of CO, although the presence of a C-C double bond is expected to partially cancel out the errors (see the trend in Table S.2 for ethane, ethylene and acetylene). Therefore, the enthalpies of C_2O_2 and related species were corrected by -0.20 eV. Our calculations indicate that CO molecules can be easily coupled through a charge-transfer mediated process to give $C_2O_2^-$. Such process has a ΔH of -0.19 eV, implying a more favorable configuration of the charged dimer (i.e. a favorable electron affinity) with respect to the separated molecules. Moreover, E_a is ~0.6 eV, which is a reasonably surmountable barrier. It is noteworthy that the addition of an extra electron decreases ΔH by ~1.5 eV and E_a by ~4 eV. Furthermore, the doubly charged anion has a ΔH of -0.31 eV, but E_a is 2.90 eV.

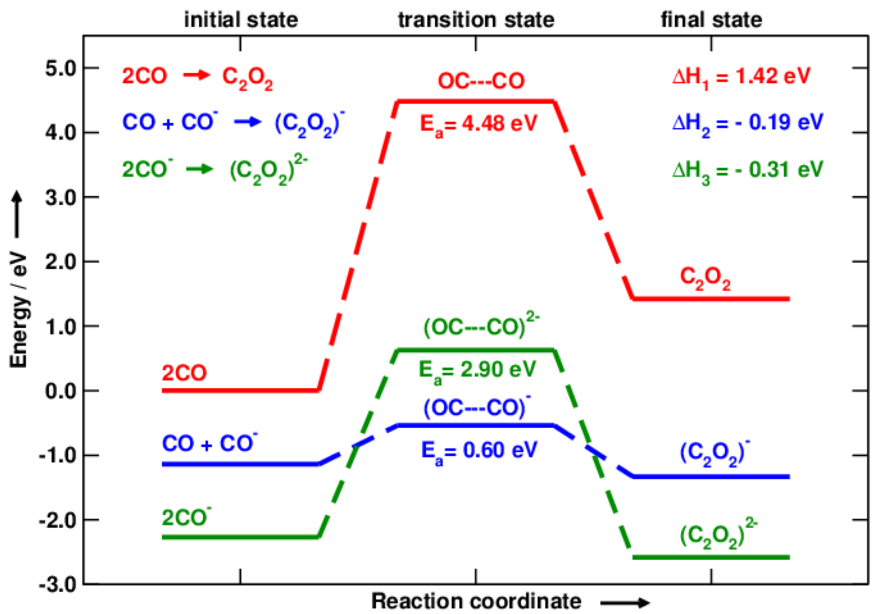

Figure S.1. Energetics of the gas-phase dimerization of CO. The initial, transition and final states are provided for the formation of C_2O_2 (red), $C_2O_2^-$ (blue) and $C_2O_2^-$ (green).

The reaction coordinate of the transition states calculated in this study is the C-C distance. Figure S.2 shows the variation of the potential energy with respect to that coordinate for neutral C_2O_2. Figures S.2, S.3 and S.4 contain only DFT-calculated data and no corrections are included. On the other hand, the barriers provided in Figure S.1 contain ZPEs and gas-phase corrections. The ZPEs of the transition states (TS) and the dimers are 0.165 eV and 0.40 eV. The equilibrium C-C distance of the dimer is 1.29 Å and the TS is found to be around 1.79 Å. The energetic difference between the initial state (IS) and the TS is 4.27 eV and the reaction is endothermic.

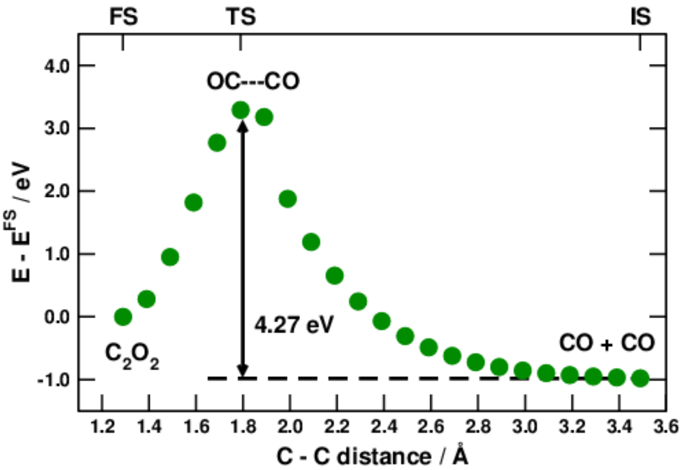

Figure S.2. Potential Energy Surface for the association of two CO molecules in vacuum. The initial, transition and final states (IS, TS, and FS, respectively) are provided in the auxiliary x-axis.

Figure S.3 shows the variation of the potential energy with respect to the reaction coordinate for $C_2O_2^-$. The equilibrium C-C distance of the charged dimer is around 1.33 Å and the transition state is found around 1.89 Å. The energetic difference between the IS (CO + CO^-) and the TS is only 0.42 eV and the reaction is exothermic.

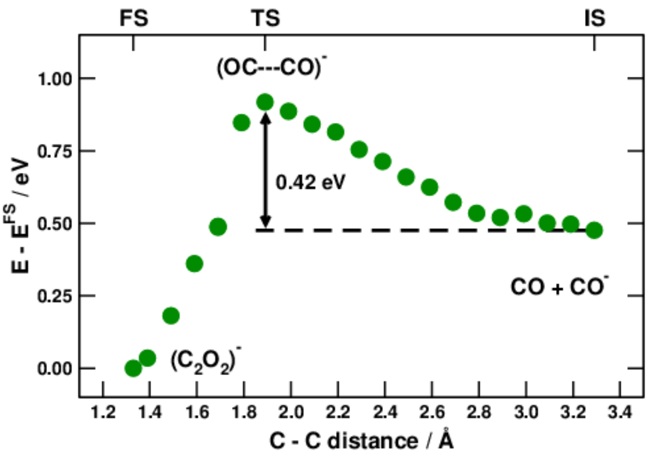

Figure S.3. Potential Energy Surface for the association of CO and CO^- in vacuum.

Figure S.4 shows the variation of the potential energy with respect to the reaction coordinate for $C_2O_2^{2-}$. The equilibrium C-C distance of the doubly charged dimer is around 1.26 Å and the transition state is found around 1.96 Å. The energetic difference between the IS (two charged CO molecules) and the TS is 2.68 eV and the reaction is exothermic.

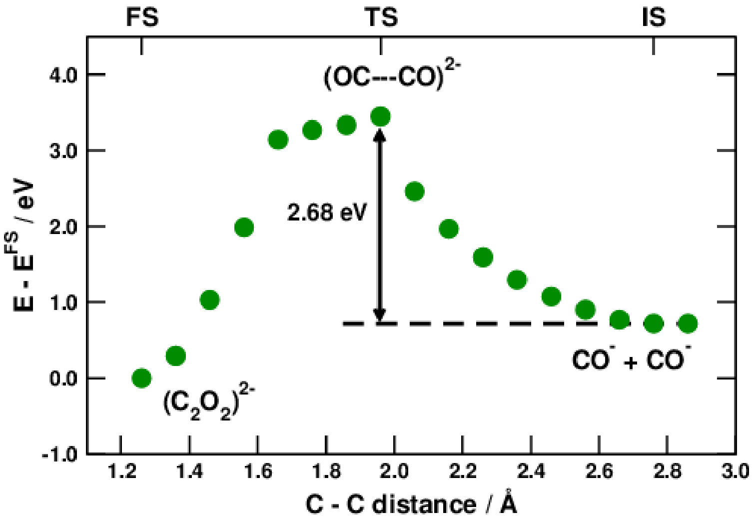

Figure S.4. Potential Energy Surface for the association of two CO^- species in vacuum.

8. Lowest-Energy Pathway for the Reduction of Ethylene oxide

The lowest-energy pathway for the reduction of ethylene oxide to ethylene is given in Figure S.5. The chemical reaction that describes the process and the potential-determining steps are given as insets. The potential-determining step is found to be the H_2O desorption from the surface, i.e. the second proton-electron transfer.

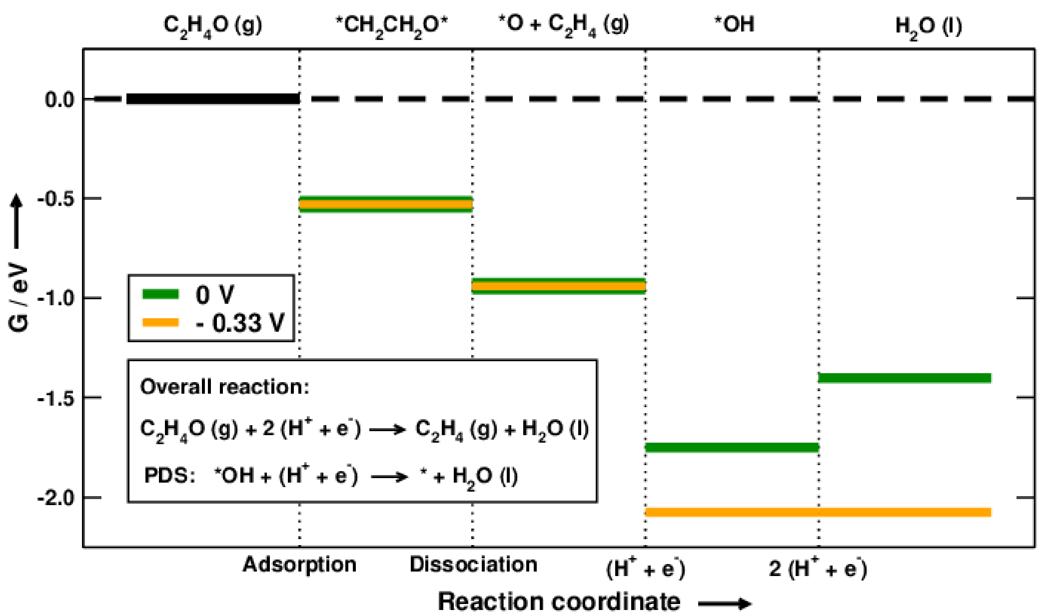

Figure S.5. Lowest-energy pathway for the electrochemical reduction of C_2H_4O to C_2H_4 at 0 V (green) and -0.33 V (orange).

9. Bader Analysis for the Transition State of the CO-CO coupling

The saddle point of the OC---CO coupling, denoted in Figure 3.a in the main text by TS, is located 0.79 eV over the gas phase reference at 0 V, approximately 0.2 eV higher than the dimer, and its imaginary frequency is around 48.5 cm^{-1}. Its ZPE and vibrational entropy are 0.37 and 0.22 eV, respectively. Moreover, we present in Figure 3.a the expected energy landscape at -0.40 V, which is the calculated onset potential. At this potential the total energy for the first proton-electron transfer is zero and the whole conversion of CO into C_2 species is expected to start. Note that the energies of two CO molecules in gas phase, *CO and 2*CO adsorbed are fixed at 0, -0.25 and -0.40 eV, whereas those of *C_2O_2 and *CO-COH are lowered by 0.40 eV. The energy of the TS is only lowered by 0.27 eV, as the total charge of the dimer at the saddle point is around 2/3 e^-, as shown below in Figure S.6. At the TS the surface donation of charge is made by more atoms than in the case of *C_2O_2 (including a Cu atom in the second layer), although the amount transferred is less.

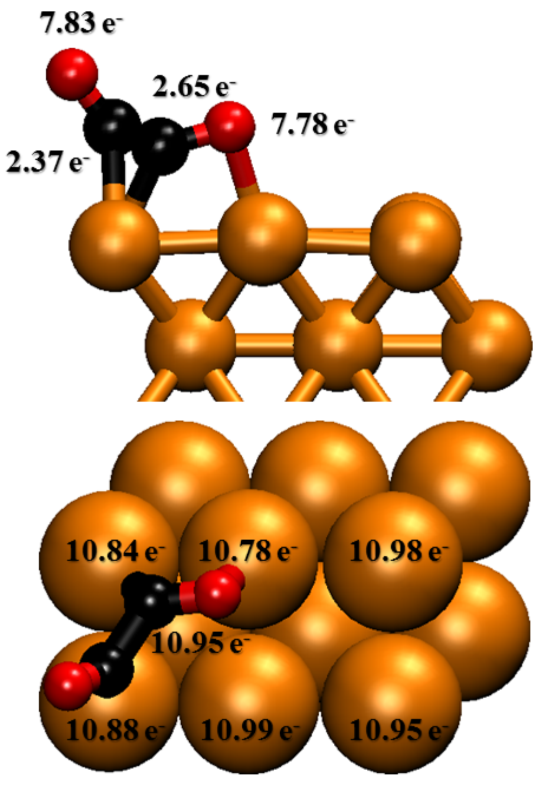

Figure S.6. Adsorption geometry and Bader charges of the TS of the CO-CO coupling on Cu(100). The lack of charge on the surface is equivalent to the excess charge on the dimer and approximately equal to 2/3 e^-.

10. Coverage and Coadsorption Effects

At the electrochemical conditions of the CO reduction, i.e. a considerably negative potential applied to a metal electrode, the surface is expected to be covered mostly with *CO and *H. Our calculations were made at a relatively high coverage of *CO as two molecules were present on the surface. The adsorption energy of the first *CO is -0.25 eV, whereas the coadsorption of two of them has a total adsorption energy of -0.40 eV. This means that the adsorption energy of *CO is reduced by ~0.075 eV/CO when *CO adsorbates are close enough to repel each other. Additionally, the presence of *H tends to slightly weaken the adsorption energies of other adsorbates, as shown in Table S.4.

Table S.4. Adsorption energies of various species with and without coadsorption of *H.

Adsorbed species	ΔG_{ADS}^{clean} / eV	ΔG_{ADS}^{H*} / eV
2*CO	-0.40	-0.33
*C_2O_2	0.56	0.59
*CO-COH	0.40	0.43

Therefore, we conclude that although *H coadsorption weakens the adsorption energies, the effect is generally small and tends to be more pronounced on *CO adsorption. This is, in fact, beneficial for the kinetics of the reaction, as it reduces the barrier between 2*CO and *C_2O_2.

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